

CORE WEB VITALS

1. LCP (Largest Contentful Paint)

- **What it measures:** How long it takes for the largest visible element (text or image) to appear on the screen.
- **Why it matters:** A slow LCP makes the site feel like it's taking forever to load.
- **Good benchmark:** Under **2.5 seconds** is considered good.
- **Examples of LCP elements:**
 - Hero image/banner
 - Main heading text
 - Featured video thumbnail
- **Optimization tips:**
 - Optimize images (compress, use WebP/AVIF)
 - Use a faster server or CDN
 - Minimize render-blocking CSS/JS

2. FID (First Input Delay)

- **What it measures:** The time from when the user first interacts (click, tap, key press) to when the browser responds.
- **Why it matters:** A high FID makes the site feel sluggish or unresponsive.
- **Good benchmark:** Less than **100 milliseconds**.
- **Common causes of poor FID:**
 - Heavy JavaScript execution blocking the main thread
 - Large third-party scripts (ads, analytics)
- **Optimization tips:**
 - Break up long JavaScript tasks
 - Use code-splitting and lazy loading
 - Reduce unused JavaScript

3. CLS (Cumulative Layout Shift)

- **What it measures:** How much the visible content unexpectedly shifts during page load.
- **Why it matters:** Sudden jumps can cause users to click the wrong button or lose their place.
- **Good benchmark:** Less than **0.1**.
- **Common causes of high CLS:**
 - Images without set width/height

- Ads or embeds resizing dynamically
- Fonts loading late causing text reflow
- **Optimization tips:**
 - Always set width and height for images/videos
 - Reserve space for ads/embeds
 - Use font-display: swap for web fonts